

Joseph McGuire

+1(805) 468-5639, joseph.c.mcg@gmail.com

Work Experience

Freelance Consultant

NextFrame Analytics (Self-Employed) — Remote (Providence, RI)

October 2024 – Present

- Delivered analytical and automation solutions for SaaS and ML-focused clients, transforming complex business needs into optimized data pipelines and dashboards.
- Queried and maintained PostgreSQL databases to analyze product performance and troubleshoot operational issues.
- Conducted cross-system data investigations integrating CloudWatch, S3, and API logs to identify inefficiencies and optimize uptime.
- Designed data flow documentation and workflow diagrams in support of business analytics and reporting needs.
- **Key Projects**
 - Font Generation Agent (SaaS): Engineered a serverless LangGraph-based pipeline enabling dynamic font creation and usage analytics, showcased at Vercel's 2025 AI Accelerator.
 - Backtesting Agent: Built an LLM-assisted financial optimization API integrating SQL-driven analysis, data validation, and scenario testing across multi-market datasets.

Technical Contributor (Part-Time)

Cognitive Chemistry Labs Inc. — Remote (Providence, RI)

January 2025 – Present

- Contribute part-time as a founding team member for an early-stage SaaS initiative applying AI to molecular simulation.
- Designed SQL schemas and data flows across PostgreSQL, AWS S3, and GPU services to support large-scale computational workloads.
- Drafted process documentation and analytics reporting templates for product and operations teams.
- Support strategic data architecture and monitoring improvements on an as-needed basis.

Research Engineer

AIMdyn Inc. — Remote (Santa Barbara, CA)

October 2022 – November 2024

- Led data pipeline design for DARPA's SCEPTER initiative, optimizing SQL and Polars workflows for high-frequency time-series analysis.
- Created detailed process and data flow documentation supporting multi-system traceability and reproducibility.
- Integrated cloud logging (AWS CloudWatch) and API monitoring into model performance reports for client delivery.
- Authored multiple data-driven technical proposals and coordinated with external research partners and national labs.
- Promoted to Principal Investigator (PI) in 2023 for leading end-to-end pipeline optimization and

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team coordination.

Data Scientist

*Brooksource (Contractor for Deloitte US - Data and Analytics) — Remote (Atascadero, CA)
June 2022 – October 2022 (6 month contract)*

- Designed and optimized secure ETL pipelines in Informatica and PostgreSQL, processing 10TB+ of sensitive financial audit data.
 - Implemented anonymization and data-quality verification routines, improving throughput by 30%.
 - Collaborated with BI teams to troubleshoot process bottlenecks and ensure compliance with internal security standards.
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Certifications

Build Intelligent Agents Using DeepSeek & n8n — Board Infinity (Issued Oct 2025) — Credential ID Q8A1LJ3F6DK0

Foundations of Project Management — Google / Coursera (Issued Sep 2025) — Credential ID EX13DNU18JN5

Technical Skills

Languages: SQL, Python (Pandas, Polars, PyTorch), JavaScript

Databases: PostgreSQL, MySQL, SQLite

Cloud & Tools: AWS (S3, CloudWatch), Docker, Heroku, GitLab CI/CD

Visualization & Documentation: Excel, Power BI, Mermaid, Lucidchart, Notion

Specialties: Data modeling, process optimization, requirements documentation, cross-system analytics, SDLC familiarity

Education

M.S. in Mathematics – *Cum Laude*

*California Polytechnic State University, San Luis Obispo
GPA: 3.8 / 4.0 – Graduated May 2022*

B.A. in Mathematics and Physics – *Cum Laude*

*Sonoma State University
GPA: 3.7 / 4.0 – Graduated May 2020*

A.A. in Computer Science

*Allan Hancock College
GPA: 3.5 / 4.0 – Graduated May 2017*